

Supplemental Tables

For: Hyun-Jung Shin, Sun Woo Nam, Heeyeon Kim et al. Postoperative delirium following dexmedetomidine versus propofol sedation in healthy older adults undergoing orthopedic lower limb surgery with spinal anesthesia: A randomized controlled trial

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Table S1. Characteristics of patients

AGENDA-ITEM	<u>Intention-to-Treat</u>		<u>Per-Protocol</u>	
	<u>Propofol</u> (n = 366)	<u>Dexmedetomidine</u> (n = 366)	<u>Propofol</u> (n = 341)	<u>Dexmedetomidine</u> (n = 342)
<u>Age (year)</u>	71 (67–75)	72 (68–76)	71 (67–76)	72 (68–76)
<u>Sex</u>				
<u>Male</u>	77 (21%)	78 (21%)	77 (23%)	70 (21%)
<u>Female</u>	289 (79%)	288 (79%)	264 (77%)	272 (79%)
<u>Weight (kg)</u>	63 (56–69)	61 (56–69)	63 (56–69)	61 (56–69)
<u>Height (cm)</u>	154 (150–159)	153 (149–159)	154 (150–159)	153 (149–159)
<u>ASA classification</u>				
<u>1</u>	66 (18%)	52 (14%)	60 (18%)	47 (14%)
<u>2</u>	300 (82%)	314 (86%)	281 (82%)	295 (86%)
<u>Underlying disease</u>				
<u>Hypertension</u>	223 (61%)	232 (64%)	205 (60%)	215 (63%)
<u>Diabetes mellitus</u>	77 (21%)	101 (28%)	70 (21%)	92 (27%)
<u>Chronic kidney disease</u>	7 (1.9%)	8 (2.2%)	5 (1.5%)	6 (1.8%)
<u>Chronic liver disease</u>	3 (0.8%)	4 (1.1%)	3 (0.9%)	2 (0.6%)
<u>Chronic pulmonary disease</u>	8 (2.2%)	7 (2.0%)	8 (2.3%)	2 (0.6%)
<u>Rheumatoid arthritis</u>	4 (1.1%)	13 (3.6%)	4 (1.2%)	10 (2.9%)

Data expressed as median (interquartile range), or n (%).

ASA, American Society of Anesthesiologists

Table S2. Perioperative surgical and anesthetic variables

EMERGENCY AMBULANCE SERVICES COMMITTEE	<u>Intention-to-Treat</u>			<u>Per-Protocol</u>		
	<u>Propofol</u> (n = 366)	<u>Dexmedetomidine</u> (n = 366)	<u>P Value</u>	<u>Propofol</u> (n = 341)	<u>Dexmedetomidine</u> (n = 342)	<u>P Value</u>
<u>Type of surgery</u>						
<u>Hip and femur</u>	92 (25%)	89 (24%)	0.667	85 (25%)	76 (22%)	0.418
<u>Knee and tibia/fibula</u>	248 (68%)	247 (68%)	0.937	231 (68%)	238 (70%)	0.621
<u>Ankle and foot</u>	26 (7%)	30 (8%)	0.472	25 (7%)	28 (8%)	0.775
<u>Surgery time (min)</u>	85 (75–105)	85 (75–105)	0.244	85 (75–105)	85 (75–105)	0.402
<u>Anesthesia time (min)</u>	125 (115–150)	130 (115–155)	0.413	125 (115–150)	130 (115–155)	0.668
<u>Estimated blood loss</u> <u>(mL)</u>	50 (50–250)	50 (50–200)	0.336	50 (50–250)	50 (50–200)	0.222
<u>Crystalloid (mL)</u>	450 (300–600)	400 (300–550)	0.258	450 (300–600)	400 (300–550)	0.173
<u>Colloid (mL)</u>	0 (0–0)	0 (0–100)	0.651	0 (0–50)	0 (0–0)	0.959
<u>Urine (mL)</u>	244 (130–400)	250 (150–380)	0.778	255 (150–414)	259 (169–398)	0.785
<u>Red blood cell (unit)</u>						
<u>Before surgery</u>	0 (0–0)	0 (0–0)	>0.999	0 (0–0)	0 (0–0)	>0.999
<u>During surgery</u>	0 (0–0)	0 (0–0)	0.799	0 (0–0)	0 (0–0)	0.804
<u>After surgery</u>	0 (0–0)	0 (0–0)	0.635	0 (0–0)	0 (0–0)	0.640
<u>Modified observer's</u> <u>assessment of</u> <u>alertness/sedation</u> <u>score at PACU</u>	5 (5–5)	5 (4–5)	<0.001	5 (5–5)	5 (4–5)	<0.001
<u>PACU stay (min)</u>	27 (20–43)	37 (23–60)	<0.001	26 (20–43)	37 (23–60)	<0.001
<u>Length of stay (day)</u>						



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<u>Before surgery</u>	<u>1 (1–1)</u>	<u>1 (1–1)</u>	<u>0.870</u>	<u>1 (1–1)</u>	<u>1 (1–1)</u>	<u>0.782</u>
<u>After surgery</u>	<u>6 (6–7)</u>	<u>6 (6–7)</u>	<u>0.778</u>	<u>6 (6–7)</u>	<u>6 (6–7)</u>	<u>0.537</u>
<u>Total</u>	<u>7 (7–8)</u>	<u>6 (6–7)</u>	<u>0.821</u>	<u>7 (7–8)</u>	<u>7 (7–8)</u>	<u>0.626</u>
<u>ICU care, postoperatively</u>	<u>1 (0.3%)</u>	<u>0 (0%)</u>	<u>>0.999</u>	<u>1 (0.3%)</u>	<u>0 (0%)</u>	<u>0.499</u>
<u>Ephedrine, total (mg)</u>	<u>5 (0–5)</u>	<u>5 (0–10)</u>	<u>0.179</u>	<u>5 (0–5)</u>	<u>5 (0–10)</u>	<u>0.181</u>
<u>Phenylephrine, total (µg)</u>	<u>0 (0–30)</u>	<u>0 (0–20)</u>	<u>0.411</u>	<u>0 (0–50)</u>	<u>0 (0–30)</u>	<u>0.223</u>
<u>Atropine, total (mg)</u>	<u>0 (0–0)</u>	<u>0 (0–0)</u>	<u>0.314</u>	<u>0 (0–0)</u>	<u>0 (0–0)</u>	<u>0.207</u>
<u>MAP (mmHg)</u>						
<u>Before sedation</u>	<u>94 (87–102 [61–122])</u>	<u>92 (83–101 [61–121])</u>	<u>0.098</u>	<u>94 (87–103 [61–125])</u>	<u>92 (84–101 [61–124])</u>	<u>0.083</u>
<u>During sedation</u>	<u>74 (69–79 [41–119])</u>	<u>77 (71–84 [53–103])</u>	<u><0.001</u>	<u>74 (69–79 [41–109])</u>	<u>77 (70–83 [53–103])</u>	<u><0.001</u>
<u>At PACU</u>	<u>80 (74–87 [60–112])</u>	<u>74 (68–80 [55–112])</u>	<u><0.001</u>	<u>79 (74–87 [60–112])</u>	<u>73 (70–80 [55–107])</u>	<u><0.001</u>
<u>HR (rates/min)</u>						
<u>Before sedation</u>	<u>74 (67–82 [46–112])</u>	<u>74 (65–82 [45–125])</u>	<u>0.194</u>	<u>74 (67–82 [46–112])</u>	<u>73 (62–83 [45–125])</u>	<u>0.130</u>
<u>During sedation</u>	<u>63 (58–70 [45–125])</u>	<u>60 (55–66 [44–104])</u>	<u><0.001</u>	<u>63 (58–70 [45–100])</u>	<u>60 (54–65 [44–104])</u>	<u><0.001</u>
<u>At PACU</u>	<u>68 (62–77 [44–100])</u>	<u>64 (58–72 [43–99])</u>	<u><0.001</u>	<u>68 (62–77 [44–99])</u>	<u>64 (58–71 [43–99])</u>	<u><0.001</u>

Data expressed as median (interquartile range), median (interquartile range [range]), or n (%).

MAP; mean arterial pressure; HR, heart rates; PACU, postanesthesia care unit; ICU, intensive care unit

Table S3. Basal cognitive status and postoperative delirium

**CHIEF
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UPDATE
REPORT**

Intention-to-Treat

Per-Protocol

	<u>Propofol</u> <u>(n = 366)</u>	<u>Dexmedetomidine</u> <u>(n = 366)</u>	<u>Odds Ratio</u> <u>(95% CI)</u>	<u>P-value</u>	<u>Propofol</u> <u>(n = 341)</u>	<u>Dexmedetomidine</u> <u>(n = 342)</u>	<u>Odds Ratio</u> <u>(95% CI)</u>	<u>P-value</u>
<u>Mini-Cog score</u>	<u>5 (3–5)</u>	<u>5 (3–5)</u>		<u>0.102</u>	<u>4 (3–5)</u>	<u>4 (3–5)</u>		<u>0.118</u>
<u>Delirium*</u>	<u>24 (6.6%)</u>	<u>11 (3.0%)</u>	<u>0.42 (0.201–0.86)</u>	<u>0.036</u>	<u>23 (6.7%)</u>	<u>10 (2.9%)</u>	<u>0.368 (0.175–0.77)</u>	<u>0.021</u>
<u>Delirium onset</u>				<u>0.802</u>				<u>0.606</u>
<u>Postoperative 1 day</u>	<u>15 (4.1%)</u>	<u>7 (1.9%)</u>			<u>14 (4.1%)</u>	<u>6 (1.8%)</u>		
<u>Postoperative 2 day</u>	<u>5 (1.4%)</u>	<u>3 (0.8%)</u>			<u>5 (1.5%)</u>	<u>3 (0.9%)</u>		
<u>Postoperative 3 day</u>	<u>4 (1.1%)</u>	<u>1 (0.3%)</u>			<u>4 (1.2%)</u>	<u>1 (0.3%)</u>		
<u>Delirium period**</u>				<u>Not applicable</u>				<u>Not applicable</u>
<u>Postoperative 1 to 2 days</u>	<u>6 (1.6%)</u>	<u>0 (0%)</u>			<u>5 (1.5%)</u>	<u>0 (0%)</u>		
<u>Postoperative 2 to 3 days</u>	<u>1 (0.3%)</u>	<u>1 (0.3%)</u>			<u>0 (0%)</u>	<u>1 (0.3%)</u>		
<u>Postoperative 1 to 3 days</u>	<u>1 (0.3%)</u>	<u>0 (0%)</u>			<u>0 (0%)</u>	<u>0 (0%)</u>		

Data expressed as median (interquartile range), n (%), or odd ratio (95% CI).

CI, confidence interval.

* For patients with missing data due to early hospital discharge, if they were evaluated as positive for the confusion assessment method even once during the postoperative admission period, they were classified as postoperative delirium: In the propofol group, 8 and 16 patients discharged at 2 and 3 days after surgery, respectively. In the dexmedetomidine group, 4 and 18 patients discharged at 2 and 3 days after surgery, respectively.

Per-Protocol population: In the propofol group, 8 and 16 patients discharged at 2 and 3 days after surgery, respectively. In the dexmedetomidine group, 3 and 17 patients discharged at 2 and 3 days after surgery, respectively.

** Only patients

Date of meeting	18/07/2023
FOI Status	Open/Public
If closed please indicate reason	Not Applicable — Public Report
Prepared by	Deputy Chief Ambulance Services Commissioner
Presented by	Chief Ambulance Services Commissioner
Approving Sponsor	Chief Ambulance Services Commissioner

Report purpose	FOR NOTING
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Engagement (internal/external) undertaken to date (including receipt/consideration at Committee/group)		
Committee/Group/Individuals	Date	Outcome
EASC Management Group	June 2023	Noted

ACRONYMS	
ASI	Ambulance Service Indicators
CASC	Chief Ambulance Services Commissioner
EMRTS	Emergency Medical Retrieval and Transfer Service
WAST	Welsh Ambulance Services NHS Trust

1. SITUATION/BACKGROUND

- 1.1—The purpose of this report is for the Members to receive an update on key matters related to the work of the Chief Ambulance Services Commissioner (CASC).
- 1.2—The content of this report was also presented to EASC Management Group in June.

~~2. — SPECIFIC MATTERS FOR CONSIDERATION BY THIS MEETING (ASSESSMENT)~~

~~2.1 — Since the last meeting progress has been made against a number of key areas which for ease of reference are listed below:~~

- ~~➤ Meetings with Welsh Ambulance Services NHS Trust (WAST)~~
- ~~➤ Review of remote clinical support~~
- ~~➤ Six Goals for Urgent and Emergency Care Programme~~
- ~~➤ Connected Support Cymru (previously known as Night Sitting Service)~~
- ~~➤ Transfer, Discharge and Repatriations~~
- ~~➤ Review of National Commissioning~~
- ~~➤ Data linking~~
- ~~➤ 1000 Beds or Equivalent Community Service.~~
- ~~➤ Health Education and Improvement Wales (HEIW) — Education commissioning of Paramedics and Advanced Paramedic Practitioners (APPs).~~

~~2.2 — Meetings with delirium lasting the Welsh Ambulance Services NHS Trust (WAST)~~

~~I have continued to hold weekly meetings with the Chief Executive of WAST and hold monthly Quality and Delivery meetings with WAST Executive Directors.~~

~~In addition to this, I also recently held a meeting with the WAST Executive Director of Operations and a number of health board operational colleagues to discuss recent changes to the use of ambulance staff for the movement of~~

patients within hospital sites and the return of patients to ambulance following diagnostics (referred to as 'portering').

Clarity was provided on the fundamentals of care and speciality definitive pathways. There was a general consensus that the focus should remain on releasing ambulances as soon as possible and WAST agreed to meet locally with HB colleagues to discuss any areas of short term flexibility and a phased approach to implementation.

2.3 ~~Review of Remote Clinical Support~~

The EASC team continue to make progress with this review in partnership with WAST colleagues. Following completion of the review, WAST colleagues will have the opportunity to review the content prior to the EASC Management Group receiving the report.

2.4 ~~Six Goals for Urgent and Emergency Care Programme~~

The Programme Board was held on 18 May 2023 and the Six Goals Programme Plan for 23/24 was discussed. Goal 1 and Goal 4 are hosted by the National Collaborative Commissioning Unit (NCCU) and the first integrated Goal 1 and 4 Board took place in June 2023. The clinical lead for Goal 4 is now in post (Tim Rogerson) and is focusing on emergency department improvement initiatives. A highlight report describing the areas of focus for Goal 4 is attached as **Appendix 1**.

2.5 — ~~'Connected Support Cymru' (previously known as Night Sitting Service)~~

~~This service is now live, Connected Support Cymru has been developed to:~~

- ~~• Improve patient safety and experience through digital and community support, particularly in times of system pressure~~
- ~~• Help patients access the care they need, schedule more than two days were included urgent care to improve efficiency & increase remote clinical consultation capabilities~~
- ~~• Create a 'Once for Wales' infrastructure to support remote clinical monitoring for routine, urgent and emergency care with 1st line clinical support that interfaces with local health board services.~~
- ~~• Support patients access the right service first time and the right bed first time for hospital care to reduce the bottle necks in emergency departments.~~

~~Ongoing evaluation of the service is taking place and EASC will receive a detailed update on this service at a future meeting.~~

2.6 — ~~Transfer, Discharge and Repatriations~~

~~WAST have continued work on the development of a proposal for the delivery of a National Discharge and Transfer model. The project to date has undertaken a review of the current system and outlined the areas of improvement and benefit.~~

~~The first iteration of the demand and capacity review of this services is currently being reviewed by WAST and the EASC team.~~

Discussions are ongoing with Betsi Cadwaladr, Hywel Dda and Swansea Bay University Health Boards on short and medium term solutions for critical care and high acuity transfer requirements.

2.7—**Review of National Commissioning**

The engagement work on this review has now been completed. We are now awaiting the findings of the review and confirmation of next steps from Welsh Government officials.

2.8—**Data Linking**

Progress with Data linking continues to develop and detailed work is ongoing between WAST and Digital Health and Care Wales (DHCW) on data standards alignment and linkage methodology.

The EAS Committee has previously requested a workshop to explore the potential outcomes of data linkage. This will now be arranged once sufficient progress has been made with the linkage methodology to support productive discussions at the workshop.

In addition to this, the EASC Team are progressing work on linkage of ambulance incident data to information within the Welsh Deprivation Index. Progress has been made and this data has now been linked, the analysis of the information has proven complex, EASC Management Group will receive the first draft analysis at the next meeting.

2.9—**1000 Beds or Equivalent Community Service (Further Faster)**

The closure report for this programme has been attached to this report for information at **Appendix 2**.

Members will note the recommendations at the end of the report and in particular the establishment of the 'Further Faster' governance structures. EASC members have requested that the EASC is engaged with the programme and explicit reference to this is made within the EASC IMTP.

2.10 Health Education and Improvement Wales (HEIW) — Education commissioning of Paramedics and Advanced Paramedic Practitioners (APPs)

Colleagues in HEIW recently hosted a stakeholder meeting around the education commissioning requirements of APP and the paramedics pipeline.

Colleagues were present from the EASC Team, Primary Care and the Six Goals for Urgent and Emergency Care Programme. It was highlighted that HEIW have been able to facilitate a significant increase in the education budget for advanced practice within WAST.

Given the implications of education commissioning on the decisions and revenue requirements of EASC, I would recommend that the Committee undertakes to invite the CEO of HEIW to a number of EASC meetings each year in order to allow the Committee to influence and inform decision making.

3. KEY RISKS/MATTERS FOR ESCALATION TO THE COMMITTEE

Members are asked to note the ongoing work in the specific areas.

4. IMPACT ASSESSMENT

Quality/Safety/Patient Experience implications	Yes (Please see detail below)
	The impact of the current performance and system pressures and the extremely high levels of handover delays will inevitably affect the patient experience including quality and safety aspects and will lead to harm to patients
Related Health and Care standard(s)	Governance, Leadership and Accountability
	All health and care standards apply.
Equality impact assessment completed	Not required
Legal implications / impact	There are no specific legal implications related to the activity outlined in this report.
Resource (Capital/ Revenue £/ Workforce) implications / Impact	Yes (Include further detail below)
	There are ongoing implications which are identified within the report
Link to Main Strategic Objective	The Committee's overarching role is to ensure its Commissioning Strategy for Emergency Ambulance Services utilising the

	five step patient pathway outlined within the National Collaborative Commissioning Quality and Delivery Agreement and the related outcomes for each care standard aligned with the Institute of Healthcare Improvement's (IHI) 'Quadruple Aim' are being progressed. This report focuses on all the above objectives, but specifically on providing strong governance and assurance.
Link to Main WBFG Act Objective	Provide high quality care as locally as possible wherever it is safe and sustainable

5. RECOMMENDATION

5.1—The Emergency Ambulance Services Committee is asked to:

- **NOTE** the information within the report
- **NOTE** the implication of the work of HEIW in relation to the education of paramedics and advanced paramedic practitioner numbers.